

Responses to Reviewers' Comments for Manuscript SIPN-01910-2025

Evolutionary Dynamics Based on Reputation in Networked Populations with Game Transitions

Addressed Comments for Publication to
IEEE Transactions on Signal and Information Processing over
Networks

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Dear Editors and Reviewers:

We are pleased to submit our revised manuscript titled “Evolutionary Dynamics Based on Reputation in Networked Populations with Game Transitions” (manuscript SIPN-01910-2025).

We greatly appreciate all comments received from the Handling Editor and the Reviewers. The manuscript has been carefully revised according to all constructive suggestions.

We would like to take this opportunity again to thank you for considering our manuscript, and we look forward to hearing from you.

Yours sincerely,

Minyu Feng, Ph.D.

On behalf of all authors

Yuji Zhang, Jürgen Kurths, and Attila Szolnoki

Note: To enhance the legibility of this response letter, all the Editor’s and reviewers’ comments are typeset in boxes.

Response to the Editor

Summary Comment

This manuscript proposes a reputation-based mechanism to enhance cooperation by linking reputation growth to the number of cooperative neighbors. While the idea is promising, the paper requires major revision before it can be considered for publication.

Response:

We are grateful for the efforts and feedback of the Editor and the three Reviewers on our manuscript. We have read the Reviewers' comments and found them very helpful to improve the presentation level of our work. We do hope that the revised version now reveals the proper value and contribution it has.

We thank the Editor for giving us the opportunity to address the critical comments. In response to the feedback provided us, we have carefully revised the manuscript to justify all specific notes or requirements. We have focused on improving clarity, hence ensuring that it now fully satisfies the expectations outlined by the Reviewers.

Comment 1 of the Editor

The background lacks clarity—particularly the role of reputation and indirect reciprocity in cooperation—which should be better motivated in the introduction.

Response:

We sincerely appreciate the Editor's insightful comments which help us to significantly improve the manuscript. To address this, we have revised the *Introduction* section to strengthen the contextual elaboration. Revisions include:

In the second paragraph, we further highlight the significant role of networks in network dynamics, not only in evolutionary game theory, but in spread of epidemics and opinions.

“...These generalized network frameworks provide the groundwork for investigations into other network dynamics, e.g., the propagation of epidemics [15], [16] and opinions [17], making them no longer limited to traditional pairwise and static interactions.”

In the third paragraph, we further review the development of theoretical research in evolutionary game theory.

“...Not long afterward, Ohtsuki *et al.* provided a simple but intuitive rule, i.e., $b/c > k$ (the ratio of the benefit b from cooperation to its corresponding cost c exceeding the average degree k), for the threshold of the dominance of cooperative behavior [20]. Notably, Allen and his colleagues provided a general solution applicable to any connected network under weak selection [21]. Then, analytical results have been obtained for multilayer networks [22], temporal networks [23], higher-order networks [24], personalized strategy updates [25], and self-interaction learning [26].”

In the fifth paragraph, we further clarify the role of stochastic games in evolutionary game theory.

“...Different from its positive impact on cooperation, varying environments can also lead to the collapse [41] and oscillations [42] of cooperation. Hence, how changing environments affect evolutionary dynamics has become the most intensively studied research path in evolutionary game theory.”

In the sixth paragraph, we further clarify the role of reputation in evolutionary game theory.

“Meanwhile, reputation serves as a pivotal mechanism fostering cooperation and overcoming social dilemmas in human societies [43], of which typical examples are image scoring [44], stern judging [45], and shunning [46]. Significant progress has been made in this field:...”

“...Hu *et al.* proposed a trust game and found that trust flourishes in networks if the reputation threshold is high [49]. Second-order reputation was considered and cooperation can be enhanced with the increase of reputation strength [50]. Cuesta *et al.* verified the unique role of reputation in fostering cooperation through human experiments [51]. In the latest work, adaptive reputation based on the snowdrift game promotes cooperation on simplicial complexes [52].”

In the seventh paragraph, we further clarify the motivations on our model.

“...Consequently, we here consider the case where the increase and decrease of the individual's reputation is reflected and determined by its neighbors' behaviors, which can be regarded as homophily in social networks [53] or neighborhood effects [54], [55], which encapsulate how human behavioral outcomes is affected by spatial contexts through interconnected social and environmental pathways.”

Please refer to Section “*I. Introduction*” for more details.

Comment 2 of the Editor

The model description also needs more detail, especially on the reputation dynamics and imitation process.

Response:

We agree with the Editor's comment on model description. Accordingly, we have revised Section “*II. Evolutionary Game In Networked Populations Based On Stochastic Games And Reputations*” and provide a more exhaustive elaboration on the proposed model.

In subsection “*A. Game Model*”, we added the following details:

“...Furthermore, the interaction pattern is determined by a networked population, in which nodes represent individuals and edges represent interactions.”

In subsection “*B. Stochastic Games*”, we added the following details:

“...To describe the changeable resources in reality, we utilize stochastic games to model them. Hence, we denote the state space $E = \{G_0, \dots, G_m\}$ as all possible $(m + 1)$ game states that can be reached during evolution.”

“...These include deterministic state-independent and probabilistic behavior-independent game transitions, of which details are listed as follows:”

“...As shown in Fig. 1(a), the current game state is G_i and there is one cooperator (the single blue node, $l = 1$), and thus the rule of game transition is determined by the elements $p_{ij}^{[1]}$ for $j \in E$ in $\mathbf{P}^{[1]}$. That is, the considered game state will change from G_i to G_j with probability $p_{ij}^{[1]}$. The difference between G_i and G_j lies in the payoff difference for mutual cooperation and unilateral defection (see Eq. 1).”

“...In particular, for the case $\mathbf{P}^{[0]} = \mathbf{P}^{[1]} = \mathbf{P}^{[2]}$, the game transition mode can be regarded as exogenous and is not influenced by strategy evolution.”

In subsection “C. Reputation Mechanism and Strategy Updating Rule”, we added the following details:

“...In general, it follows the cognition that if one individual has more cooperators around, then it enjoys a higher reputation, and vice versa. This neighbor-dependent reputation mechanism effectively reflects the “neighborhood effects”, widely existing in societies.”

“...Concerning the imitation process, different from the random choice from neighbors, it is a natural assumption that individuals are more likely to imitate those whose fitness F is higher.”

“...Concretely, if $\delta = 0$ (resp. $\delta = 1$), the updating process is driven by payoff difference $P_x(t) - P_y(t)$ (resp. reputation difference $R_x(t) - R_y(t)$).”

Please refer to Section “II. Evolutionary Game In Networked Populations Based On Stochastic Games And Reputations” for more details.

Comment 3 of the Editor

Lastly, the paper presentation should be improved considerably for clarity.

Response:

We fully agree that improving the presentation is crucial for enhancing the readability and impact of our work, and we therefore made the following changes.

In the revised version, we have made substantial efforts to refine the paper’s presentation, including:

(1): Clearly express the setting for all parameters and the rationale behind them. We also provide the parameter values for each simulation. Revisions are:

We clarify the settings for parameters that appear frequently in this paper and the rationales behind them. Revisions are

“...For the setting of the payoff matrix in Eq. 1, as usual, we always set $c = 1$ by default and vary b_i to describe the dilemma strength and effect of game transitions.

For better presentation of results, we adopt different values of b_i in different simulations, and the concrete value is clarified in each simulation. In this work, unless otherwise stated, we let $a = 4$ in Eq. 6 and $\kappa = 1$ and $\delta = 0.2$ in Eq. 8, which account for a suitable relation between $R_i(t)$ and $F_i(t)$, a payoff-reputation weighted strategy updating rule, and a moderate selection strength with a certain noise.”

“...Concerning the parameters related to the reputation mechanism in Eq. 5, for the details of the variation of reputation ξ_{n_c} , we set $\xi_0 \sim N(-0.2, 0.1)$, $\xi_1 \sim N(0, 0.1)$, $\xi_2 \sim N(0.1, 0.1)$, and for $n \geq 3$, $\xi_n \sim N(0.3, 0.1)$ to avoid the excessively rapid and excessively slow growth in reputation, and the initial value of reputation is fixed to $R(0)$, making the coefficient in Eq. 6 equals 1.”

For Fig. 2 in the latest manuscript, the added information in each caption about parameters is

“...Other parameters are: $c = 1$, $\delta = 0.2$, $a = 4$, and $B_1 = 4$ under SL and $B_1 = 3$ under WS.”

For Figs. 4-5 in the latest manuscript, the added information in each caption about parameters is

“...Other parameters are: $c = 1$, $\delta = 0.2$, $a = 4$, and $b_1 = 2$.”

For Fig. 6 in the latest manuscript, the added information in the caption about parameters is

“...Other parameters: $b_0 = 2.7$ and $b_1 = 2$ under SL, $b_0 = 3.6$ and $b_1 = 3$ under WS, $c = 1$, $\delta = 0.2$, $\kappa = 1$, and $a = 4$.”

For Fig. 7 in the latest manuscript, the added information in the caption about parameters is

“...Other parameters are: $c = 1$, $\delta = 0.2$, $\kappa = 1$, and $a = 4$.”

For Fig. 8 in the latest manuscript, the added information in the caption about parameters is

“...Other parameters are: $b_0 = 3.7$ under SL and $b_1 = 3.65$ under WS, $b_1 = 3$, $c = 1$, and $\kappa = 1$.”

For Figs. 9 and 10 in the latest manuscript, the added information in the caption about parameters is

“...Other parameters are: $c = 1$, $\delta = 0.2$, $\kappa = 1$, and $a = 4$.”

For Fig. 11 in the latest manuscript, the added information in the caption about parameters is

“...Other parameters are the same as those in Fig. 2.”

For Fig. 12 and 13 in the latest manuscript, the added information in the caption about parameters is

“...Other parameters are: $b_0 = 4.5$, $b_1 = 3$, $c = 1$, $\delta = 0.2$, $\kappa = 1$, and $a = 4$.”

(2): Modify confusing statements and enhance readability. Revisions are:

We have changed the expression “Singularity of interaction” to “Uniformity of interactions” in Fig. 1(c) and provided a detailed description.

“...Subplots (b) and (c) show the game transitions on networked populations, where each player has two strategies to choose from, i.e., cooperation (blue) and defection (red), and the edge color indicates the game state. Subplots (b) and (c) exhibit interactive relationships’ diversity and uniformity, respectively. The diversity indicates that each individual can face different social dilemmas at the same time (depicted by different colors in (b)). In contrast, the uniformity means all individuals face the same social dilemma at the same time (depicted by the same color in (c)), and the social dilemma changes as time progresses.”

We have modified the sentence “Furthermore, the heatmap does not show a left-skewed or right-skewed trend in both networks.” to

“Two heatmaps both exhibit distinct vertical strip-like patterns, in which the color variations are primarily confined to changes along the δ . This highlights the important role of δ and reputation in evolutionary outcomes.”

(3): We have provided the relevant simulation result for the phenomenon described in Fig. 3. The result is in line with the statements in the main text: if we make this change, cooperators fail to entirely dominate the considered network (see the Inset in Fig. R1).

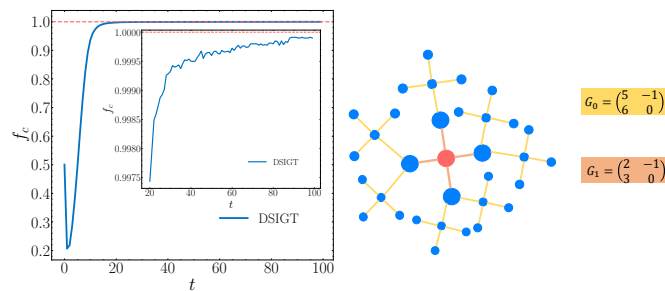


Fig. R1: Illustration of the particular dilemma. The center player is the defector and the others cooperate. This condition is expected to persist for an extended period, especially under strong selection. Parameters are the same as those in Fig. 2 during simulations.

We believe these revisions will significantly enhance the overall clarity of the paper. Thank you again for your insightful suggestions, which help us to improve our work.

Response to Reviewer 1

Summary Comment

In this manuscript, the authors studied the evolution of cooperation under the combined effects of reputation and game transitions. In the model, the reputation is linked to neighbors' strategies, while the game transition may depend on both individual decisions and game environment, or only be determined by external environments. Numerical simulation results are presented, which offer potential insights into collective cooperation dynamics in temporally varying environments. However, there are several issues that should be addressed, and the following are the comments in detail.

Response:

We thank our Reviewer for his/her careful reading, thoughtful review, and constructive feedback. We have carefully revised the manuscript according to these comments and suggestions. Please find our specific responses as follows.

Comment 1.1

In the model, the authors assumed that reputation is determined by neighbors' strategies. However, in real-world scenarios, reputation typically emerges from individuals' own behaviors rather than being defined by peers. Could the authors provide empirical evidence from sociological studies or concrete examples for this neighbor-dependent reputation assumption?

Response 1.1:

Our proposed neighbor-dependent reputation mechanism shares the same spirit of "neighborhood effects". Neighborhood effects encapsulate how spatial contexts shape human outcomes through interconnected social, economic, and environmental pathways. From this perspective, our proposed model focuses on the impact of "neighborhood reputation" on evolutionary dynamics, which is scientifically reasonable and has wide practical applications.

References

- [1] Helms A C. Keeping up with the Joneses: Neighborhood effects in housing renovation[J]. *Regional Science and Urban Economics*, 2012, 42(1-2): 303-313.
- [2] Evans M, Lee B A. Neighborhood reputations as symbolic and stratifying mechanisms in the urban hierarchy[J]. *Sociology Compass*, 2020, 14(10): 1-15.
- [3] Tran E, Blankenship K, Whittaker S, et al. My neighborhood has a good reputation: Associations between spatial stigma and health[J]. *Health & Place*, 2020, 64: 102392.
- [4] Permentier M, Van Ham M, Bolt G. Neighbourhood reputation and the intention to leave the neighbourhood[J]. *Environment and Planning A*, 2009, 41(9): 2162-2180.

To support the mechanism of our model, we have added relevant descriptions about neighbor effects in Section *Introduction*. Details are (the mentioned Refs. [54] and [55] are exactly References [1] and [3] listed above):

“...Consequently, we here consider the case where the increase and decrease of the individual’s reputation is reflected and determined by its neighbors’ behaviors, which can be regarded as homophily in social networks [53] or neighborhood effects [54], [55], which encapsulate how human behavioral outcomes is affected by spatial contexts through interconnected social and environmental pathways.”

Comment 1.2

What is the meaning of “Singularity of interaction” in Fig. 1(c)? A brief description is expected.

Response 1.2:

We thank our Reviewer for careful reading and insightful comments. The term “Singularity of interaction” means all individuals interact with their neighbors in the same social dilemma (represented by the same color). For comparison, in Fig. 2(b), one individual can face different social dilemmas (represented by different colors). To avoid confusion, we have changed it to “Uniformity of interactions” for better elucidation.

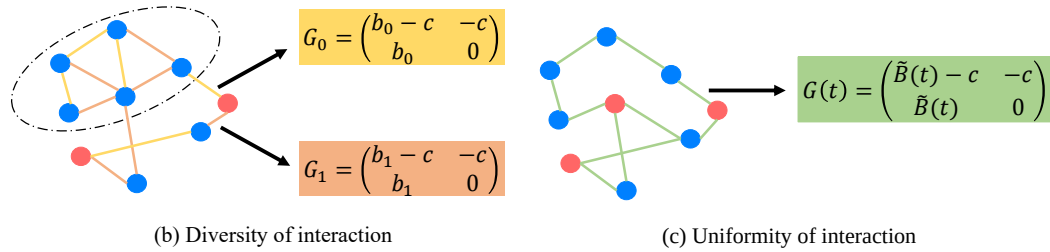


Fig. R2: “...Subplots (b) and (c) show the game transitions on networked populations, where each player has two strategies to choose from, i.e., cooperation (blue) and defection (red), and the edge color indicates the game state. Subplots (b) and (c) exhibit interactive relationships’ diversity and uniformity, respectively. The diversity indicates that each individual can face different social dilemmas at the same time (depicted by different colors in (b)). In contrast, the uniformity means all individuals face the same social dilemma at the same time (depicted by the same color in (c)), and the social dilemma changes as time progresses...”

Please refer to Fig. 1 in the revised manuscript for more details.

Comment 1.3

As shown in Fig.2, the evolution outcomes are insensitive to the initial values of GBM in SL, but sensitive in WS. Why? Could the authors provide an explanation?

Response 1.3:

We appreciate your constructive comment. The inconsistency comes from the topology difference. It is well recognized that evolutionary dynamics can present diverse outcomes, determined by specific structures. Square lattice (SL) with periodic boundary conditions, for instance, represents a regular graph where each individual has the same number of neighbors;

while in a Watts–Strogatz network (WS) nodes may have different degrees and have the well-known small-world property. The homogeneity eliminates the impact of the initial values of GBM, and therefore, a slight rise is observed even when the initial value of GBM is increased. We have added relevant explanations in the main text:

“...The insensitivity observed in SL is partly caused by its regular topological structure, which eliminates the impact of the initial values of GBM.”

Please refer to the revised manuscript for more details.

Comment 1.4

How does the selection strength, i.e., κ , affect the results? The parameter κ is fixed at 1 throughout the manuscript. As it may significantly affect the outcomes, the authors should provide more simulation results for larger or smaller values of κ , or a brief discussion.

Response 1.4:

We sincerely appreciate your constructive recommendation. Inspired by it, in the revised manuscript, we have provided the numerical analysis on how the noise parameter κ affects the evolution of cooperation (see Fig. R3). Details are listed as follows:

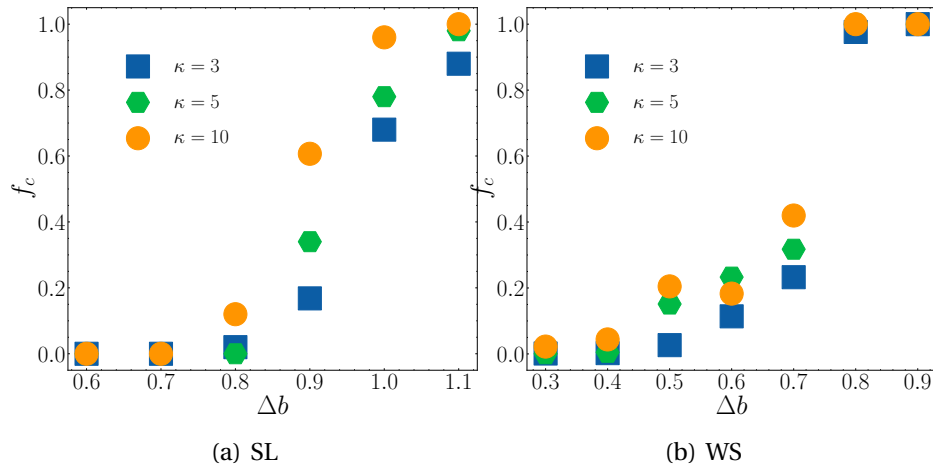


Fig. R3: Plots of cooperation frequency under different noises κ . The ranges of x -axis are set as $[0.6, 1.1]$ and $[0.3, 0.9]$ in subplots (a) and (b), with steps equal to 0.1, whereas the ranges of y -axis in two subplots are set as $[0.0, 1.0]$. b_0 is fixed as 3 in both subplots. Other parameters are: $c = 1$, $\delta = 0.2$, $a = 4$, and $b_1 = 2$.

“...In this section, we remove the restrictions on the noise parameter $\kappa = 1$ to explore how the evolution of cooperation is affected by noise strengths. In the following numerical analysis, we set three values of κ , i.e., $\kappa = 3$, $\kappa = 5$, $\kappa = 10$, and corresponding results are exhibited in Fig. 5.

According to Fig. 5 (SL in Fig. 5(a) and WS in Fig. 5(b)), we can observe that under larger noise κ , it is more beneficial for the evolution of cooperation, especially

in SL. This finding indicates that κ plays a positive role in promoting cooperation. It also verifies the robustness of our model.”

Please refer to Section “D. Evolution under Different Noises” for more details.

Comment 1.5

On page 5, “Interestingly, if we change the update rule, i.e., Eq. 8 from the difference between two individuals’ payoff to the difference between two individuals’ fitness...”. According to Eq. 6, fitness depends on both payoff and reputation. The expression on the right side of Eq. 8 includes two terms, related to the difference in payoff and the difference in reputation, respectively. How to change the rule in Eq. 8 from the difference between payoff to the difference between fitness?

Response 1.5:

Thank you for your insightful observation. We appreciate for pointing out this ambiguity in our description of Eq. 8, and we apologize for the unclear statement. Recall that Eq. 8 is

$$\mathbb{P}(s_x \leftarrow s_y) = \frac{1 - \delta}{1 + \exp\{[P_x(t) - P_y(t)]/\kappa\}} + \frac{\delta}{1 + \exp\{[R_x(t) - R_y(t)]/\kappa\}},$$

where $P_x(t)$ and $R_x(t)$ are the payoff and reputation of individual x at time t respectively. The mentioned change is to modify this equation to

$$\mathbb{P}(s_x \leftarrow s_y) = \frac{1 - \delta}{1 + \exp\{[F_x(t) - F_y(t)]/\kappa\}} + \frac{\delta}{1 + \exp\{[R_x(t) - R_y(t)]/\kappa\}}.$$

where $F_x(t)$ is the fitness of individual x at time t .

As pointed out our Reviewer, $F_x(t)$ depends on both payoff and reputation, and this change would therefore lead to the repetition of $R_x(t)$ in Fermi updating. As a consequence, we just provided a brief discussion on this case and mainly persisted Eq. 8 in the article.

Comment 1.6

On page 7, “Furthermore, the heatmap does not show a left-skewed or right-skewed trend in both networks.” What is the meaning of this sentence? Please clarify it.

Response 1.6:

Thanks for your invaluable suggestion, and we sincerely apologize for the unclear original sentence. We have clarified it now and the revised version is:

“...Furthermore, two heatmaps both exhibit distinct vertical strip-like patterns, in which the color variations are primarily confined to changes along the δ . This highlights the important role of δ and reputation in evolutionary outcomes.”

Please refer to Section “G. Cooperation Frequency on the (δ, a) Parameter Space” for more details.

Comment 1.7

There are some typos. For examples, - at the end of page 7, "...where G_0 and G_3 are the most and least valuable states respectively", should be G_0 and G_2 . On the next page, "...tends to result in G_0 (resp. G_3).", should be "...tends to result in G_0 (resp. G_2). - on page 5, "we make a preliminary discussion on PBIGI" should be "...PBIGT".

Response 1.7:

Thanks for your careful reading again! We have corrected all the mentioned typos. For the first typo, the corresponding sentence has been rewritten to

"...Therefore, we now turn to game transitions among three states (G_0 , G_1 , and G_2), where G_0 and G_2 are the most and least valuable states respectively, i.e., $b_0 > b_1 > b_2$."

"...Eq. 11 indicates that the game transitions from a deterministic to a probabilistic pattern, where mutual cooperation (resp. mutual defection) tends to result in G_0 (resp. G_2)."

For the second typo, the corresponding sentence has been rewritten to

"...Therefore, we make a preliminary discussion on PBIGT and in the following subsections and focus on the endogenous game transitions."

Last, we wish to thank our Reviewer again for his/her efforts in reviewing our manuscript. These invaluable comments have not simply helped us to improve the clarity of our presentation, but these comments were also stimulating to extend our results significantly.

Response to Reviewer 2

Summary Comment

In this manuscript, the authors proposed a type of growth mechanism of reputation reflected by the cooperation level among its neighbors, which is determined by the number of cooperators. It is recognized that cooperation survives by forming clusters to mitigate the exploitation from defectors. Meanwhile, through comprehensive and intensive simulations, we have uncovered that increasing reputation strength δ during the imitation process benefits the cooperative behavior, but the reputation-fitness parameter a does not work significantly. Taking together, the current idea could be interesting, and it can be potentially accepted after some minor revisions as follows

Response:

We thank Reviewer 2 for evaluating our manuscript and the overall positive assessment. We are grateful for your insightful and constructive suggestions which have significantly helped us to improve our manuscript. Based on your feedback, we have made detailed revisions and enhancements, which are outlined below the corresponding comments. Our specific responses are the following.

Comment 2.1

The background is not very clear, and the authors had better highlight it in the introduction. In particular, regarding the role of reputation or indirect reciprocity in the evolution of cooperation, the authors should further clarify it;

Response 2.1:

We sincerely appreciate this comment. In response, we have revised the *Introduction* section to strengthen the contextual elaboration. Details are:

In the second paragraph, we further highlight the significant role of networks in network dynamics, not only in evolutionary game theory, but in spread of epidemics and opinions.

“...These generalized network frameworks provide the groundwork for investigations into other network dynamics, e.g., the propagation of epidemics [15], [16] and opinions [17], making them no longer limited to traditional pairwise and static interactions.”

In the third paragraph, we further review the development of theoretical research in evolutionary game theory.

“...Not long afterward, Ohtsuki *et al.* provided a simple but intuitive rule, i.e., $b/c > k$ (the ratio of the benefit b from cooperation to its corresponding cost c exceeding the average degree k), for the threshold of the dominance of cooperative behavior [20]. Notably, Allen and his colleagues provided a general solution applicable to any connected network under weak selection [21]. Then, analytical results have been obtained for multilayer networks [22], temporal networks [23], higher-order networks [24], personalized strategy updates [25], and self-interaction

learning [26].”

In the fifth paragraph, we further clarify the role of stochastic games in evolutionary game theory.

“...Different from its positive impact on cooperation, varying environments can also lead to the collapse [41] and oscillations [42] of cooperation. Hence, how changing environments affect evolutionary dynamics has become one of the main topics in evolutionary game theory.”

In the sixth paragraph, we further clarify the role of reputation in evolutionary game theory.

“Meanwhile, reputation serves as a pivotal mechanism fostering cooperation and overcoming social dilemmas in human societies [43], of which typical examples are image scoring [44], stern judging [45], and shunning [46]. Significant progress has been made in this field:...”

“...Hu *et al.* proposed a trust game and found that trust flourishes in networks if the reputation threshold is high [49]. Second-order reputation was considered and cooperation can be enhanced with the increase of reputation strength [50]. Cuesta *et al.* verified the unique role of reputation in fostering cooperation through human experiments [51]. In the latest work, adaptive reputation based on the snowdrift game promotes cooperation on simplicial complexes [52].”

In the seventh paragraph, we further clarify the motivations on our model.

“...Consequently, we here consider the case where the increase and decrease of the individual's reputation is reflected and determined by its neighbors' behaviors, which can be regarded as homophily in social networks [53] or neighborhood effects [54], [55], which encapsulate how human behavioral outcomes is affected by spatial contexts through interconnected social and environmental pathways.”

Please refer to Section “*I. Introduction*” for more details.

Comment 2.2

The model can be depicted in detail;

Response 2.2:

We can agree with this suggestion. Therefore, we have revised Section “*II. Evolutionary Game In Networked Populations Based On Stochastic Games And Reputations*” and provide a more exhaustive elaboration on the proposed model.

In subsection “*A. Game Model*”, we added the following details:

“...Furthermore, the interaction pattern is determined by a networked population, in which nodes represent individuals and edges represent interactions.”

In subsection “*B. Stochastic Games*”, we added the following details:

“...To describe the changeable resources in reality, we utilize stochastic games to model them. Hence, we denote the state space $E = \{G_0, \dots, G_m\}$ as all possible $(m + 1)$ game states that can be reached during evolution.”

“...These include deterministic state-independent and probabilistic behavior-independent game transitions, of which details are listed as follows:”

“...As shown in Fig. 1(a), the current game state is G_i and there is one cooperator (the single blue node, $l = 1$), and thus the rule of game transition is determined by the elements $p_{ij}^{[1]}$ for $j \in E$ in $\mathbf{P}^{[1]}$. That is, the considered game state will change from G_i to G_j with probability $p_{ij}^{[1]}$. The difference between G_i and G_j lies in the payoff difference for mutual cooperation and unilateral defection (see Eq. 1).”

“...In particular, for the case $\mathbf{P}^{[0]} = \mathbf{P}^{[1]} = \mathbf{P}^{[2]}$, the game transition mode can be regarded as exogenous and is not influenced by strategy evolution.”

In subsection “C. Reputation Mechanism and Strategy Updating Rule”, we added the following details:

“...In general, it follows the cognition that if one individual has more cooperators around, then it enjoys a higher reputation, and vice versa. This neighbor-dependent reputation mechanism effectively reflects the “neighborhood effects”, widely existing in societies.”

“...Concerning the imitation process, different from the random choice from neighbors, it is a natural assumption that individuals are more likely to imitate those whose fitness F is higher.”

“...Concretely, if $\delta = 0$ (resp. $\delta = 1$), the updating process is driven by payoff difference $P_x(t) - P_y(t)$ (resp. reputation difference $R_x(t) - R_y(t)$).”

Please refer to Section “II. Evolutionary Game In Networked Populations Based On Stochastic Games And Reputations” for more details.

Comment 2.3

Regarding the role of hybrid update between the payoff and reputation, Eq.(8) uses the same noise parameter κ , I suggest the authors try different ones since they apply into different scenarios.

Response 2.3:

We sincerely appreciate your constructive recommendation. Inspired by it, in the revised manuscript, we have provided a numerical analysis on how the noise parameter κ affects the evolution of cooperation (see Fig. R4). Details are listed as follows:

“...In this section, we remove the restrictions on the noise parameter $\kappa = 1$ to explore how the evolution of cooperation is affected by noise strengths. In the following numerical analysis, we set three values of κ , i.e., $\kappa = 3$, $\kappa = 5$, $\kappa = 10$, and corresponding results are exhibited in Fig. 5.

According to Fig. 5 (SL in Fig. 5(a) and WS in Fig. 5(b)), we can observe that under larger noise κ , it is more beneficial for the evolution of cooperation, especially

in SL. This finding indicates that κ plays a positive role in promoting cooperation. It also verifies the robustness of our model.”

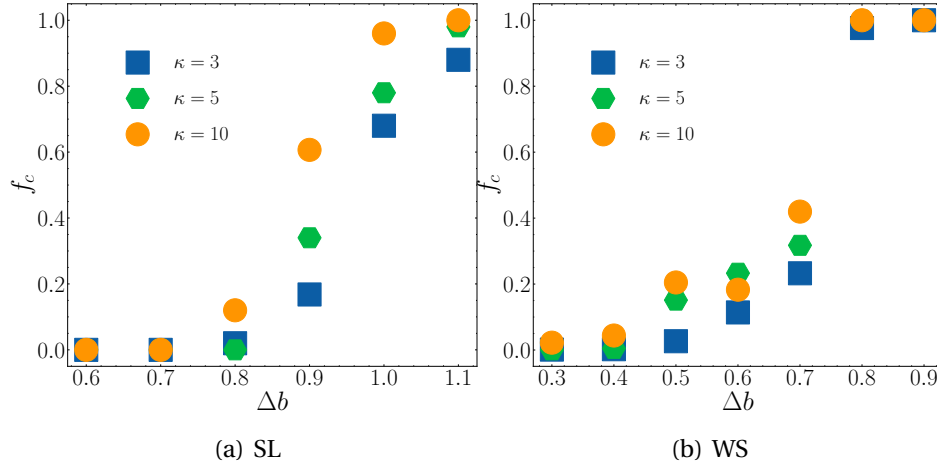


Fig. R4: Plots of cooperation frequency under different noises κ . The ranges of x -axis are set as $[0.6, 1.1]$ and $[0.3, 0.9]$ in subplots (a) and (b), with steps equal to 0.1, whereas the ranges of y -axis in two subplots are set as $[0.0, 1.0]$. b_0 is fixed as 3 in both subplots. Other parameters are: $c = 1$, $\delta = 0.2$, $a = 4$, and $b_1 = 2$.

Please refer to Section “D. Evolution under Different Noises” for more details.

Comment 2.4

All tables should be redrawn in three-line style.

Response 2.4:

It is an excellent idea!. We have carefully revised all tables in the manuscript to follow the three-line style as requested. We present the revised table below.

Table 1: Results of the Frequency of the Whole Population Evolves into Pure States

b_0	b_1	simulations	pure cooperators	pure defectors
2.5	1.1	1000	938	62
2.6	1.1	1000	987	13
2.7	1.1	1000	996	4
2.8	1.1	1000	998	2

Last, we wish to thank our Reviewer again for his/her efforts in reviewing our manuscript. These invaluable comments have helped us to significantly improve the clarity of our manuscript.

Response to Reviewer 3

Summary Comment

This paper examines how game dynamics and reputation influence the evolution of cooperation within the framework of the donation game. The study investigates the outcomes associated with this evolutionary process, considering various topologies for distinct interactions. Additionally, a biased mutation is introduced to provide insights into strategy evolution. Simulations reveal an increase in cooperation levels, and several important phenomena are observed. However, there are several points that the authors should address.

Response:

We thank Reviewer 3 for reviewing our manuscript, and we are sincerely grateful for your insightful comments and recommendations, which have significantly helped us to improve our manuscript. Guided by your feedback, we have made detailed revisions and interpretations, which are outlined below the corresponding comments. Our specific responses are listed below.

Comment 3.1

In the strategy update equation (8), the reputation strength (δ) is defined to range from 0 to 1, with a value of 0.2 utilized in the simulation experiments. The results presented in Figure 7 show that the range of (δ) is set between [0.15, 0.26]. Could you please provide the results for (δ) within the full range of 0 to 1?

Response 3.1:

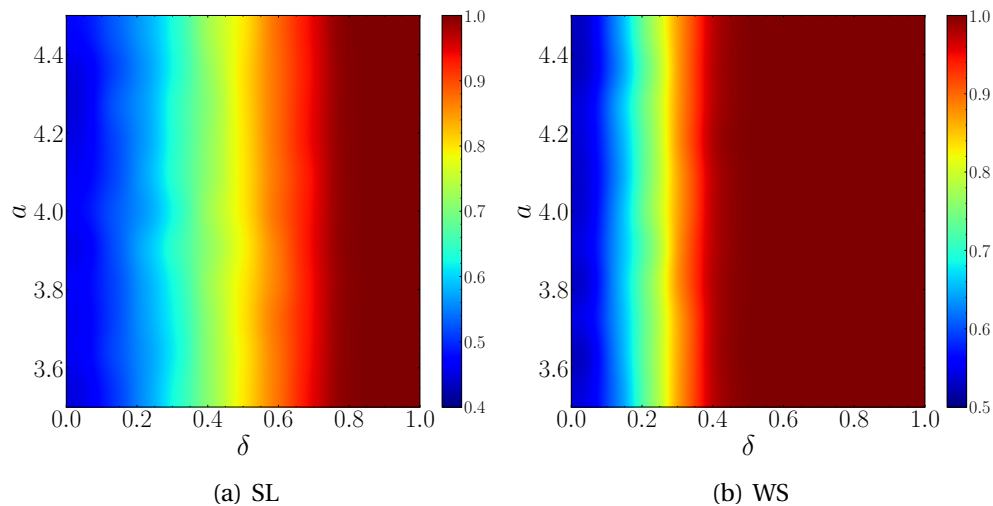


Fig. R5: Heatmaps of cooperation frequency (f_c) on δ - a parameter plane in SL (panel (a)) and WS (panel (b)) networks. The x -axis is set as δ within the range [0.0, 1.0] and y -axis is set as a within the range [3.5, 4.5]. The meaning of colors is explained in the right-hand side legend for each panel. The outcomes are averaged over 50 independent simulations to maintain a good accuracy of the simulation results. Other parameters are: $b_0 = 3.7$ under SL and $b_1 = 3.65$ under WS, $b_1 = 3$, $c = 1$, and $\kappa = 1$.

We appreciate this constructive comment. Inspired by this, we have provided the required result for δ within the full range of 0 to 1 in the latest manuscript. The revised figures are given in Fig. R5.

Notably, our core conclusions remain robust across the full range. For better illustration, we have revised part of the descriptions in the main text.

“...Furthermore, two heatmaps both exhibit distinct vertical strip-like patterns, in which the color variations are primarily confined to changes along the δ . This highlights the important role of δ and reputation in evolutionary outcomes.”

Please refer to Section “G. Cooperation Frequency on the (δ, a) Parameter Space” for more details.

We appreciate your attention to this detail, which has strengthened the rigor of our analysis. The improved figures and corresponding text have been updated accordingly in the revised manuscript.

Comment 3.2

Page 8 states, “...if we change the update rule, i.e., Eq. 8 from the difference between two individuals’ payoff to the difference between two individuals’ fitness...” Could you provide the relevant simulation results? Additionally, please explain the rationale behind the definition of fitness in equation (6). What would the results be if reputation $R_i(t)$ were directly mapped to $(0, a)$ instead of being multiplied by the payoff $P_i(t)$ as it currently is?

Response 3.2:

We thank Reviewer 3 for these critical comments.

For comment 1:

We provide the relevant simulation results in Fig. R6. The result is in line with the statements in the main text: if we make this change, cooperators fail to entirely dominate the considered network (see the Inset in Fig. R6).

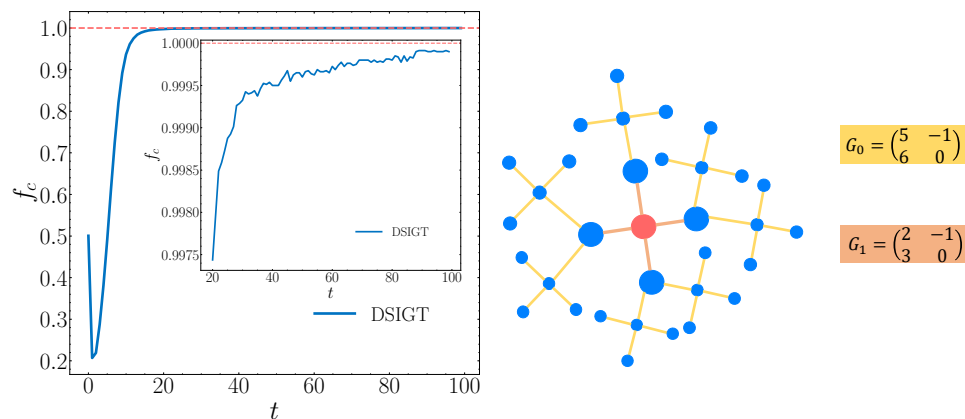


Fig. R6: Illustration of the particular dilemma. The center player is the defector, and the others cooperate. This condition is expected to persist for an extended period, especially under strong selection. Parameters are the same as those in Fig. 2 during simulations.

We have updated Fig. 3 in the revised manuscript to better present the mentioned result. Please refer to the revised manuscript for more details.

For comment 2:

(1) It is natural to assume that fitness is shaped by the combined effects of reputation and payoff, as these are two crucial elements in evolutionary dynamics. Since we allow $R_i(t)$ to take any real value ($R_i(t) \in \mathbb{R}$), we must impose restrictions on how reputation influences fitness—after all, in reality, reputation would not exert an unbounded impact on an individual's fitness.

To this end, we use the function $\frac{a}{\pi} \arctan[R_i(t)] + \frac{a}{2}$ to map $R_i(t)$ from $\mathbb{R}$ to the interval $(0, a)$. This choice stems from the fact that $\arctan(x)$ has a range of $(-\pi/2, \pi/2)$. Scaling by $\frac{a}{\pi}$ shifts this range to $(-a/2, a/2)$ and adding the final term $a/2$ then adjusts it to $(0, a)$, as required.

To capture the combined effects, we thus define fitness as the product of $\frac{a}{\pi} \arctan[R_i(t)] + \frac{a}{2}$ and $P_i(t)$. These considerations collectively motivate the definition in Eq. 6.

(2) We present the results, in which reputation is directly mapped to $(0, a)$ in Fig. R7. As exhibited in Fig. R7, it has an overall advantage in promoting cooperation in SL and WS. Concretely, it can accelerate the evolution to pure cooperators.

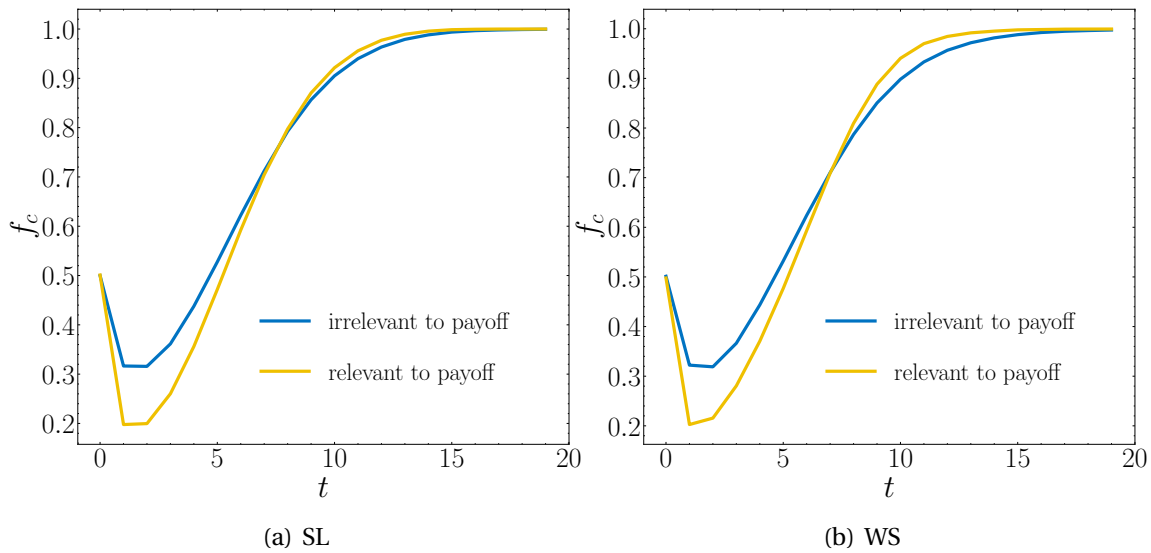


Fig. R7: Illustrations on how the rules of reputation affect the evolution of cooperation. We consider two different rules: one is $F_i(t) = \{\frac{a}{\pi} \arctan[R_i(t)] + \frac{a}{2}\} \cdot P_i(t)$ (relevant to payoff) and another is $F_i(t) = \frac{a}{\pi} \arctan[R_i(t)] + \frac{a}{2}$ (irrelevant to payoff). Parameters are: $c = 1$, $\delta = 0.2$, $a = 4$, $b_0 = 6$, and $b_1 = 3$.

Comment 3.3

The paper provides a considerable number of parameter settings, and it is suggested that the authors clearly articulate the settings for each parameter and the rationale behind them. Furthermore, the figure captions should include information about the parameter values whenever possible.

Response 3.3:

We highly agree with this suggestion. Inspired by it, in Section “A. Methods”, we clarify the settings for parameters that appear frequently in this paper and the rationales behind them. Details are

“...For the setting of the payoff matrix in Eq. 1, as usual, we always set $c = 1$ by default and vary b_i to describe the dilemma strength and effect of game transitions. For better presentation of results, we adopt different values of b_i in different simulations, and the concrete value is clarified in each simulation. In this work, unless otherwise stated, we let $a = 4$ in Eq. 6 and $\kappa = 1$ and $\delta = 0.2$ in Eq. 8, which account for a suitable relation between $R_i(t)$ and $F_i(t)$, a payoff-reputation weighted strategy updating rule, and a moderate selection strength with a certain noise.”

“...Concerning the parameters related to the reputation mechanism in Eq. 5, for the details of the variation of reputation ξ_{n_c} , we set $\xi_0 \sim N(-0.2, 0.1)$, $\xi_1 \sim N(0, 0.1)$, $\xi_2 \sim N(0.1, 0.1)$, and for $n \geq 3$, $\xi_n \sim N(0.3, 0.1)$ to avoid the excessively rapid and excessively slow growth in reputation, and the initial value of reputation is fixed to $R(0)$, making the coefficient in Eq. 6 equals 1.”

Please refer to Section “A. Methods” for more details.

Furthermore, in each figure caption, we have provided the information about the parameters:

For Fig. 2 in the latest manuscript, the added information in each caption about parameters is

“...Other parameters are: $c = 1$, $\delta = 0.2$, $a = 4$, and $B_1 = 4$ under SL and $B_1 = 3$ under WS.”

For Fig. 3 in the latest manuscript, the added information in each caption about parameters is

“...Parameters are the same as DSIGTs in Fig. 2 during simulations.”

For Figs. 4-5 in the latest manuscript, the added information in each caption about parameters is

“...Other parameters are: $c = 1$, $\delta = 0.2$, $a = 4$, and $b_1 = 2$.”

For Fig. 6 in the latest manuscript, the added information in the caption about parameters is

“...Other parameters: $b_0 = 2.7$ and $b_1 = 2$ under SL, $b_0 = 3.6$ and $b_1 = 3$ under WS, $c = 1$, $\delta = 0.2$, $\kappa = 1$, and $a = 4$.”

For Fig. 7 in the latest manuscript, the added information in the caption about parameters is

“...Other parameters are: $c = 1$, $\delta = 0.2$, $\kappa = 1$, and $a = 4$.”

For Fig. 8 in the latest manuscript, the added information in the caption about parameters is

“...Other parameters are: $b_0 = 3.7$ under SL and $b_1 = 3.65$ under WS, $b_1 = 3$, $c = 1$, and $\kappa = 1$.”

For Figs. 9 and 10 in the latest manuscript, the added information in the caption about parameters is

“...Other parameters are: $c = 1$, $\delta = 0.2$, $\kappa = 1$, and $a = 4$.”

For Fig. 11 in the latest manuscript, the added information in the caption about parameters is

“...Other parameters are the same as those in Fig. 2.”

For Fig. 12 and 13 in the latest manuscript, the added information in the caption about parameters is

“...Other parameters are: $b_0 = 4.5$, $b_1 = 3$, $c = 1$, $\delta = 0.2$, $\kappa = 1$, and $a = 4$.”

Please refer to the latest manuscript for more details.

Finally, we wish to thank our Reviewer again for his/her efforts in reviewing our manuscript. These invaluable comments have helped us to significantly improve the clarity of our manuscript.